

Lie Theory for the Impatient

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For my dearest dear, Hui.

Abstract

This document collects a variety of definitions and results from Lie theory, roughly following Kirillov's book. Proofs are only sketched briefly, sometimes insultingly so. Occasionally, motivating remarks are included, but they are kept to a minimum. The hurried reader may skip all remarks and sketches.

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1 Lie Algebras

All vector spaces in this section are complex. Before starting, let's overview where we are going.

- Section 1.1 provides the basic definitions of the study of Lie algebras, such as Lie algebras and their representations.
- Section 1.2 provides some facts about an “easy” subclass of Lie algebras. In short, nilpotent and solvable Lie algebras are those which are amenable to study by inductive arguments.
- Section 1.3 begins the study of the remaining Lie algebras, which are the semisimple ones.
- Sections 1.4 to 1.6 classify semisimple Lie algebras.
- Sections 1.7 and 1.8 study the representation theory of semisimple Lie algebras.

1.1 Starting Out

Definition 1 (Lie algebra). A Lie algebra is a finite-dimensional vector space $\mathfrak{g}$ equipped with a Lie bracket $[-, -]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying the following properties.

- (i) Bilinear: the map $[-, -]$ is bilinear.
- (ii) Alternating: $[X, X] = 0$ for all $X \in \mathfrak{g}$.
- (iii) Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ for all $X, Y, Z \in \mathfrak{g}$.

A morphism of Lie algebras is a linear map preserving the bracket.

Example 2. For any (possibly non-commutative, non-unital) algebra A over $\mathbb{C}$, one can define the Lie bracket $[-, -]: A \times A \rightarrow A$ by

$$[a, b] := ab - ba.$$

For example, applying this to $A = \text{End}(V)$ where V is a vector space gives the Lie algebra $\mathfrak{gl}(V)$. If further $V = \mathbb{C}^n$, we abbreviate $\mathfrak{gl}(V)$ to $\mathfrak{gl}_n(\mathbb{C})$.

Example 3. The subspace

$$\mathfrak{sl}_n(\mathbb{C}) := \{X \in \mathfrak{gl}_n(\mathbb{C}) : \text{tr } X = 0\}$$

is a Lie subalgebra.

Example 4. The subspace of alternating matrices is a Lie subalgebra, called $\mathfrak{o}_n(\mathbb{C}) \subseteq \mathfrak{gl}_n(\mathbb{C})$.

Remark 5 (Ado). A rather basic question in the theory is if any Lie algebra $\mathfrak{g}$ can be embedded into $\mathfrak{gl}_n(\mathbb{C})$ for some $n \geq 1$. This turns out to be a rather difficult theorem due to Ado. Let's describe one approach, which uses ideas from later subsections: use Theorem 45 to handle the semisimple and solvable cases separately. The semisimple case uses the adjoint representation, and the solvable case is handled inductively.

Definition 6 (representation). A representation of a Lie algebra $\mathfrak{g}$ is a vector space V along with a morphism $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$. A morphism of representations is a morphism of the underlying vector spaces which preserves the $\mathfrak{g}$ -action.

Example 7 (adjoint). The adjoint representation of a Lie algebra $\mathfrak{g}$ is the morphism $\text{ad}_\bullet: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ defined by

$$\text{ad}_X(Y) := [X, Y].$$

The fact that $\text{ad}_\bullet$ preserves the Lie bracket is equivalent to the Jacobi identity.

Example 8. For any representation V of $\mathfrak{g}$, there is a dual representation $V^\vee$ whose underlying vector space is the dual, and $X \in \mathfrak{g}$ acts on $\ell: V \rightarrow \mathbb{C}$ by $(X\ell)(v) := -\ell(Xv)$.

Example 9. One can take sums of representations. For two representations V and W , one can also take a tensor product: $X \in \mathfrak{g}$ acts by $X(v \otimes w) = Xv \otimes w + v \otimes Xw$.

Definition 10 (irreducible). A representation $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is *irreducible* if and only if $V \neq 0$, and V admits no nonzero proper subrepresentations. A representation is *completely reducible* if and only if it is a direct sum of irreducible representations.

Example 11. Let $\mathfrak{b} \subseteq \mathfrak{gl}_2(\mathbb{C})$ be the Lie subalgebra of upper-triangular matrices. Then the representation $\mathfrak{b} \rightarrow \mathfrak{gl}_2(\mathbb{C})$ is not completely reducible.

Lemma 12 (Schur). Let $\rho_V: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ and $\rho_W: \mathfrak{g} \rightarrow \mathfrak{gl}(W)$ be representations, and let $\varphi: V \rightarrow W$ be a nonzero morphism of representations.

- (a) If V is irreducible, then φ is injective.
- (b) If W is irreducible, then φ is surjective.
- (c) If both V and W are irreducible, then φ is an isomorphism.

Sketch. Here, (c) follows from (a) and (b). For (a), because φ is nonzero, $\ker \varphi$ is a proper subrepresentation of V , so $\ker \varphi = 0$ because V is irreducible. The proof of (b) is similar. ■

Example 13. Let the natural action of $\mathfrak{sl}_2(\mathbb{C})$ on $\mathbb{C}^2$ be irreducible. It turns out that the irreducible representations of $\mathfrak{sl}_2(\mathbb{C})$ are exactly the symmetric powers of $\mathbb{C}^2$. To see this, show that $\text{diag}(+1, -1)$ acts semisimply on any irreducible representation, and then one can parameterize the irreducible representations by the highest eigenvalue of $\text{diag}(+1, -1)$.

Definition 14 (universal enveloping algebra). Fix a Lie algebra $\mathfrak{g}$. Then the *universal enveloping algebra* $U\mathfrak{g}$ of $\mathfrak{g}$ is the free complex algebra generated by $\mathfrak{g}$ along with the relations

$$XY - YX = [X, Y]$$

for each $X, Y \in \mathfrak{g}$.

Remark 15. The data of a representation $\mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is equivalent to the data of a ring homomorphism $U\mathfrak{g} \rightarrow \text{End}(V)$.

Theorem 16 (Poincaré–Birkhoff–Witt). Fix a Lie algebra $\mathfrak{g}$, and let $\{X_i\}_{i=1}^n$ be an (ordered) basis. Then the monomials

$$\{X_{i_1} \cdots X_{i_m} : i_1 \leq i_2 \leq \cdots \leq i_m\}$$

form a basis of $U\mathfrak{g}$.

Sketch. Certainly $U\mathfrak{g}$ is spanned by monomials, and one can use the relations $XY = YX + [X, Y]$ in order to rearrange the entries of the monomial. Thus, the given monomials span. Checking that they are linearly independent amounts to a long and technical argument. ■

1.2 Nilpotent and Solvable Lie Algebras

Definition 17 (ideal). Fix a Lie algebra $\mathfrak{g}$. Then an *ideal* $I \subseteq \mathfrak{g}$ is a subspace such that $[X, I] \subseteq I$ for any $X \in \mathfrak{g}$. Equivalently, I is a $\mathfrak{g}$ -subrepresentation of $\mathfrak{g}$.

Example 18 (center). Fix a Lie algebra $\mathfrak{g}$. Then the *center* is the subspace

$$\mathfrak{z}(\mathfrak{g}) := \{X \in \mathfrak{g} : [X, \mathfrak{g}] = 0\}.$$

The center is an ideal. It is kernel the adjoint representation $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$.

Example 19 (commutator). Fix a Lie algebra $\mathfrak{g}$. Then the *commutator* is the subspace $[\mathfrak{g}, \mathfrak{g}]$. By the Jacobi identity, $[\mathfrak{g}, \mathfrak{g}]$ is a Lie ideal.

Remark 20. If I is an ideal of a Lie algebra $\mathfrak{g}$, the quotient space $\mathfrak{g}/I$ is a Lie algebra, and there is a short exact sequence

$$0 \rightarrow I \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}/I \rightarrow 0$$

of $\mathfrak{g}$ -modules.

Definition 21. A Lie algebra $\mathfrak{g}$ is *abelian* if and only if $[\mathfrak{g}, \mathfrak{g}] = 0$.

Example 22. For any Lie algebra, the quotient $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian.

Definition 23 (nilpotent). A Lie algebra $\mathfrak{g}$ is *nilpotent* if and only if there is a sequence

$$\mathfrak{g} = \mathfrak{a}_0 \supseteq \mathfrak{a}_1 \supseteq \cdots \supseteq \mathfrak{a}_k = 0$$

such that $\mathfrak{a}_{i+1}$ is an ideal of $\mathfrak{a}_i$ with abelian quotient.

Remark 24. It is equivalent to check if the sequence $\mathfrak{a}_{i+1} := [\mathfrak{g}, \mathfrak{a}_i]$ eventually stabilizes at 0.

Example 25. The subspace of strictly upper-triangular matrices of $\mathfrak{gl}_n(\mathbb{C})$ is nilpotent.

Definition 26 (solvable). A Lie algebra $\mathfrak{g}$ is solvable if and only if there is a sequence

$$\mathfrak{g} = \mathfrak{a}_0 \supseteq \mathfrak{a}_1 \supseteq \cdots \supseteq \mathfrak{a}_k = 0$$

such that each $\mathfrak{a}_i$ is an ideal of $\mathfrak{g}$ such that $\mathfrak{g}/\mathfrak{a}_i$ is abelian.

Remark 27. It is equivalent to check if the sequence $\mathfrak{a}_i := [\mathfrak{a}_{i+1}, \mathfrak{a}_{i+1}]$ eventually stabilizes at 0.

Example 28. Any nilpotent Lie algebra is solvable.

Example 29. The subspace of upper-triangular matrices of $\mathfrak{gl}_n(\mathbb{C})$ is solvable but not nilpotent.

Theorem 30 (Lie). Let $\mathfrak{g}$ be a solvable Lie algebra. For any representation V of $\mathfrak{g}$, there is a basis in which $\mathfrak{g}$ acts by upper-triangular matrices.

Sketch. By induction on $\dim V$, it is enough to show that the action of $\mathfrak{g}$ on V admits a nonzero eigenvector. For this, we induct on $\dim \mathfrak{g}$. By solvability, one may find a codimension-1 ideal $\mathfrak{g}' \subseteq \mathfrak{g}$; choose any $X \in \mathfrak{g} \setminus \mathfrak{g}'$. Now, induction tells us that V admits a nonzero eigenvector v for $\mathfrak{g}'$. With some difficulty, can show that

$$W := \text{span}_{\mathbb{C}} X^{\mathbb{N}} v$$

is a $\mathfrak{g}$ -subrepresentation of V , and it turns out that $\mathfrak{g}'$ acts diagonally on all of W . Thus, any eigenvector of X acting on W will do. ■

Corollary 31. Let $\mathfrak{g}$ be a Lie algebra. Then $\mathfrak{g}$ is solvable if and only if $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.

Sketch. If $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent, then one can show that $\mathfrak{g}$ is solvable by building ideals from the short exact sequence

$$0 \rightarrow [\mathfrak{g}, \mathfrak{g}] \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \rightarrow 0.$$

Conversely, if $\mathfrak{g}$ is solvable, then by Theorem 30, one can find a basis of $\mathfrak{g}$ on which the adjoint action of $\mathfrak{g}$ on $\mathfrak{g}$ is given by upper-triangular matrices. Thus, the action of $[\mathfrak{g}, \mathfrak{g}]$ is given by strictly upper-triangular matrices. This is enough to show $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent. ■

Remark 32. Similarly, if $\mathfrak{g}$ is nilpotent, one can show that any representation V admits a basis on which $\mathfrak{g}$ acts by strictly upper-triangular matrices.

Theorem 33 (Engel). Let $\mathfrak{g}$ be a Lie algebra. Then $\mathfrak{g}$ is nilpotent if and only if the operator $\text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ is nilpotent for all $X \in \mathfrak{g}$.

Sketch. If ad_X is nilpotent for all $X \in \mathfrak{g}$, then $[\mathfrak{g}, [\mathfrak{g}, [\mathfrak{g}, \dots, [\mathfrak{g}, \mathfrak{g}]]]] = 0$ for a high enough nesting level, so $\mathfrak{g}$ is nilpotent. Conversely, if $\mathfrak{g}$ is nilpotent, then Remark 32 grants a basis of $\mathfrak{g}$ upon which $\mathfrak{g}$ acts by strictly upper-triangular matrices, so ad_X is nilpotent for all $X \in \mathfrak{g}$. ■

1.3 Simple and Semisimple Lie Algebras

Definition 34 (simple). A Lie algebra $\mathfrak{g}$ is *simple* if and only if it is not abelian and admits no nonzero proper ideals.

Example 35. One can check by hand that $\mathfrak{sl}_2(\mathbb{C})$ is a simple Lie algebra.

Definition 36 (semisimple). A Lie algebra $\mathfrak{g}$ is *semisimple* if and only if it admits no nonzero solvable ideals.

Example 37. Any simple Lie algebra is semisimple. We will have many more examples shortly.

Remark 38. It turns out that any semisimple Lie algebra $\mathfrak{g}$ is a direct sum of simple ones. In short, one can decompose the adjoint action of $\mathfrak{g}$ on $\mathfrak{g}$ into a sum of irreducible subrepresentations, and each irreducible subrepresentation is a simple Lie subalgebra.

Remark 39. If $\mathfrak{g}$ is semisimple, then $\mathfrak{z}(\mathfrak{g}) = 0$, so the adjoint representation $\mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is injective.

Definition 40 (reductive). A Lie algebra $\mathfrak{g}$ is *reductive* if and only if $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ is semisimple.

Example 41. Because $\mathfrak{z}(\mathfrak{gl}_2(\mathbb{C}))$ consists of the scalar matrices, the Lie algebra $\mathfrak{gl}_2(\mathbb{C})$ is reductive.

Remark 42. A reductive Lie algebra $\mathfrak{g}$ is semisimple if and only if $\mathfrak{z}(\mathfrak{g}) = 0$.

Definition 43 (radical). Fix a Lie algebra $\mathfrak{g}$. Then the *radical* $\text{rad } \mathfrak{g}$ is the sum of all solvable ideals of $\mathfrak{g}$. One can check that $\text{rad } \mathfrak{g}$ is a solvable ideal.

Example 44. For a Lie algebra $\mathfrak{g}$, we see $\text{rad } \mathfrak{g} = 0$ if and only if $\mathfrak{g}$ is semisimple. For example, $\mathfrak{g}/\text{rad } \mathfrak{g}$ is semisimple.

Theorem 45 (Levi decomposition). Fix any Lie algebra $\mathfrak{g}$. Then there is a semisimple Lie subalgebra $\mathfrak{g}_{\text{ss}} \subseteq \mathfrak{g}$ and a direct sum decomposition

$$\mathfrak{g} = \text{rad } \mathfrak{g} \oplus \mathfrak{g}_{\text{ss}}.$$

Sketch. This is a very difficult result. The idea is to show that the short exact sequence

$$0 \rightarrow \text{rad } \mathfrak{g} \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}_{\text{ss}} \rightarrow 0$$

admits a splitting. Such an extension problem reduces to a calculation of some cohomology groups, which is tricky. ■

Proposition 46. Let V be a representation of a Lie algebra $\mathfrak{g}$. Suppose that the pairing $\langle -, - \rangle_V : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ defined by

$$\langle X, Y \rangle_V := \text{tr}(XY; V)$$

is non-degenerate. Then $\mathfrak{g}$ is reductive.

Sketch. It is enough to show that any $X \in [\mathfrak{g}, \text{rad } \mathfrak{g}]$ lives in the kernel of $\langle -, - \rangle_V$ (and thus vanishes). By passing to some short exact sequences, we may assume that V is irreducible. Using Theorem 30, one can show that each $\text{rad}(\mathfrak{g})$ -eigenspace of V is $\mathfrak{g}$ -invariant, so irreducibility implies that $\text{rad } \mathfrak{g}$ acts by scalars on V . A direct calculation now shows that $[\mathfrak{g}, \text{rad } \mathfrak{g}]$ acts by zero on V . ■

Example 47. The standard representation $V = \mathbb{C}^n$ of $\mathfrak{gl}_n(\mathbb{C})$ satisfies the hypothesis of Proposition 46, so $\mathfrak{gl}_n(\mathbb{C})$ is reductive. A similar argument holds for $\mathfrak{so}_n(\mathbb{C})$.

Theorem 48 (Cartan). Let $\mathfrak{g}$ be a semisimple Lie algebra. Then the pairing

$$\langle X, Y \rangle_{\mathfrak{g}} := \text{tr}(XY; \mathfrak{g})$$

is non-degenerate. This pairing is called the *Killing form*.

Sketch. Let $\mathfrak{h}$ be the kernel of $\langle -, - \rangle_{\mathfrak{g}}$, which is an ideal. We will show that $\mathfrak{h}$ is a solvable, which forces $\mathfrak{h} = 0$. It is enough to show that $\text{ad}(\mathfrak{h}) \subseteq \mathfrak{gl}(\mathfrak{h})$ is solvable. Some linear algebra shows that the pairing $\langle X, Y \rangle := \text{tr}(XY; \mathfrak{h})$ on $\mathfrak{h}$ vanishes. Thus, we are in the situation that we have a Lie algebra $\mathfrak{h}'$ with a faithful representation V for which

$$\text{tr}(XY; V) = 0$$

for each $X, Y \in \mathfrak{h}'$. One can use the Jordan decomposition to show that any element of $[\mathfrak{h}', \mathfrak{h}']$ is nilpotent, so $[\mathfrak{h}', \mathfrak{h}']$ is nilpotent, so $\mathfrak{h}'$ is solvable. ■

1.4 The Root Decomposition

Definition 49. Fix a Lie algebra $\mathfrak{g}$. Then $X \in \mathfrak{g}$ is *nilpotent* or *semisimple* if and only if $\text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ is nilpotent or semisimple, respectively.

Proposition 50 (Jordan decomposition). Fix a semisimple Lie algebra $\mathfrak{g}$. For any $X \in \mathfrak{g}$, there is a unique semisimple $X_s \in \mathfrak{g}$ and $X_n \in \mathfrak{g}$ such that $[X_s, X_n] = 0$ and $X = X_s + X_n$.

Sketch. The usual Jordan decomposition produces a unique pair $(X'_s, X'_n) \in \mathfrak{gl}(\mathfrak{g})^2$ for which X'_s is semisimple, X'_n is nilpotent, $[X'_s, X'_n] = 0$, and $\text{ad}_X = X'_s + X'_n$. Thus, it remains to check that X'_s and X'_n are in the image of $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$. In short, the Jordan decomposition shows that X'_s and X'_n are polynomials in ad_X , so they are derivations of $\mathfrak{g}$, but all derivations of $\mathfrak{g}$ come from $\mathfrak{g}$. ■

Definition 51 (Cartan). Fix a semisimple Lie algebra $\mathfrak{g}$. Then a *torus* is an abelian subalgebra $\mathfrak{h}$ of $\mathfrak{g}$ consisting of semisimple elements. A maximal torus is a *Cartan subalgebra*.

Example 52. The subalgebra of diagonal matrices of $\mathfrak{sl}_n(\mathbb{C})$ is a Cartan subalgebra.

Remark 53. Let $\mathfrak{h}$ be a Cartan subalgebra. It turns out that

$$\mathfrak{h} = \{X \in \mathfrak{g} : [X, \mathfrak{h}] = 0\}.$$

In short, use the Jordan decomposition and the maximality of $\mathfrak{h}$.

Remark 54. It turns out that any two Cartan subalgebras of $\mathfrak{g}$ are conjugate under $\text{Aut } \mathfrak{g}$. This is fairly technical, and we will only use it to show that certain constructions are independent of the choice of $\mathfrak{h}$.

Definition 55 (rank). Fix a semisimple Lie algebra $\mathfrak{g}$. Then the *rank* of $\mathfrak{g}$ is the dimension of a Cartan subalgebra of $\mathfrak{g}$.

Example 56. The rank of $\mathfrak{sl}_n(\mathbb{C})$ is $n - 1$.

Definition 57. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Then the *root system* $\Phi \subseteq \mathfrak{h}^*$ is the set of nonzero eigenspaces of $\mathfrak{h}$ acting on $\mathfrak{g}$.

Proposition 58. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Then the *root decomposition* is the eigenspace decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_{\alpha},$$

where $\mathfrak{g}_{\alpha}$ is the eigenspace of $\alpha \in \Phi$.

Sketch. Diagonalize the representation $\mathfrak{g}$ of $\mathfrak{h}$. ■

Remark 59. For $\alpha, \beta \in \Phi$, we have $[\mathfrak{g}_{\alpha}, \mathfrak{g}_{\beta}] \subseteq \mathfrak{g}_{\alpha+\beta}$.

Remark 60. The subspace Φ spans $\mathfrak{h}^*$. Indeed, any $h \in \mathfrak{h}$ orthogonal to all $\alpha \in \Phi$ then has $\text{ad}_h = 0$, so $h \in \mathfrak{z}(\mathfrak{g})$, so $h = 0$.

Notation 61. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$ and root system $\Phi \subseteq \mathfrak{h}^*$. For any $\alpha \in \Phi$, the subspaces $\mathfrak{g}_{\alpha}$ and $\mathfrak{g}_{-\alpha}$ generate a subalgebra of $\mathfrak{g}$ isomorphic to $\mathfrak{sl}_2(\mathbb{C})$. (This is a little tricky to show.) We denote this subalgebra by $\mathfrak{sl}_2(\mathbb{C})_{\alpha}$.

Proposition 62. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$ and root system $\Phi \subseteq \mathfrak{h}^*$. Let $(-, -)$ be the Killing form on $\mathfrak{g}$.

- (a) For each $\alpha \in \Phi$, $\dim \mathfrak{g}_{\alpha} = 1$.
- (b) For each $\alpha, \beta \in \Phi$, $2(\alpha, \beta)/(\alpha, \alpha) \in \mathbb{Z}$.
- (c) For each $\alpha \in \Phi$, define the reflection $s_{\alpha} : \mathfrak{h}^* \rightarrow \mathfrak{h}^*$ defined by $s_{\alpha}(\lambda) := \lambda - 2(\alpha, \lambda)/(\alpha, \alpha) \cdot \alpha$. Then s_{α} preserves Φ .
- (d) If $\alpha \in \Phi$ and $k\alpha \in \Phi$ for some $k \in \mathbb{C}$, then $k \in \{+1, -1\}$.

Sketch. Let $(-, -)$ be the Killing form on $\mathfrak{h}^*$, which is non-degenerate. Define $h_\alpha := 2H_\alpha/(\alpha, \alpha)$, where H_α is the element dual to $\alpha \in \mathfrak{h}^*$. Then one can show that

$$\mathbb{C}h_\alpha \oplus \bigoplus_{k \in \mathbb{Z} \setminus \{0\}} \mathfrak{g}_{k\alpha}$$

is an irreducible representation of $\mathfrak{sl}_2(\mathbb{C})_\alpha$. Each part now follows from understanding the irreducible representations of $\mathfrak{sl}_2(\mathbb{C})$. ■

Example 63. Set $\mathfrak{g} := \mathfrak{sl}_n(\mathbb{C})$, and let $\mathfrak{h}$ be the diagonal Cartan subalgebra. For each $i \in \{1, 2, \dots, n\}$, let $e_i : \mathfrak{h} \rightarrow \mathbb{C}$ be the i th projection. For each distinct i and j , the elementary matrix E_{ij} is an eigenvector for the adjoint $\mathfrak{h}$ -action with eigenvalue $e_i - e_j$. Thus, $\Phi = \{e_i - e_j : i \neq j\}$. Technically, $\mathfrak{h}^*$ should be thought of as $\mathbb{C}^n/\mathbb{C}(e_1 + \dots + e_n)$.

Remark 64. One can recover $\mathfrak{g}$ from the root system $\Phi \subseteq \mathfrak{h}^*$. Indeed, $\mathfrak{g}$ has basis given by a basis of $\mathfrak{h}^*$ and a vector e_α for each $\alpha \in \Phi$. Roughly speaking, the Lie bracket is determined by requiring $[e_\alpha, e_\beta] \in \mathbb{C}e_{\alpha+\beta}$ via the $\mathfrak{sl}_2(\mathbb{C})_\alpha$ -representation generated by e_β . A more precise version of this remark is given by the Serre presentation of a semisimple Lie algebra, which we will not discuss further.

1.5 Root Systems

Definition 65 (reflection). A Euclidean vector space E is a finite-dimensional real vector space equipped with an inner product $\langle -, - \rangle$. Fix $\alpha \in E$. Then define the reflection $s_\alpha \in O(E)$ as

$$s_\alpha(\lambda) := \lambda - \frac{2(\alpha, \lambda)}{(\alpha, \alpha)} \cdot \alpha$$

Definition 66 (root system). Fix a Euclidean vector space E . A root system is a nonempty finite subset $\Phi \subseteq E \setminus \{0\}$ such that

- (a) Φ spans E ;
- (b) for each $\alpha, \beta \in \Phi$, the number $n_{\alpha\beta} := 2(\alpha, \beta)/(\beta, \beta)$ is an integer; and
- (c) Φ is stable under s_α for each $\alpha \in \Phi$.

The root system Φ is *reduced* if and only if $\mathbb{Z}\alpha \cap \Phi = \{\pm\alpha\}$ for each $\alpha \in \Phi$. The *rank* of a root system is $\dim_{\mathbb{R}} E$.

Remark 67. By Cauchy-Schwarz, $n_{\alpha\beta} \in \{-2, -1, -1/2, 0, +1/2, +1, +2\}$.

Example 68. For any semisimple Lie algebra $\mathfrak{g}$ with Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Then the root system $\Phi \subseteq \mathfrak{h}^*$ is a reduced root system. The root system of $\mathfrak{sl}_n(\mathbb{C})$ is denoted A_{n-1} .

Example 69. If $\Phi_1 \subseteq E_1$ and $\Phi_2 \subseteq E_2$ are root systems, then $\Phi_1 \sqcup \Phi_2 \subseteq E_1 \times E_2$ is also a root system. Here, $\Phi_1 \subseteq E_1 \times E_2$ refers to the subset $\{(\alpha, 0) \in E_1 \times E_2 : \alpha \in \Phi_1\}$.

Definition 70 (coroot). Fix a Euclidean vector space E . For $\alpha \in E$, its coroot $\alpha^\vee \in E^*$ is defined by

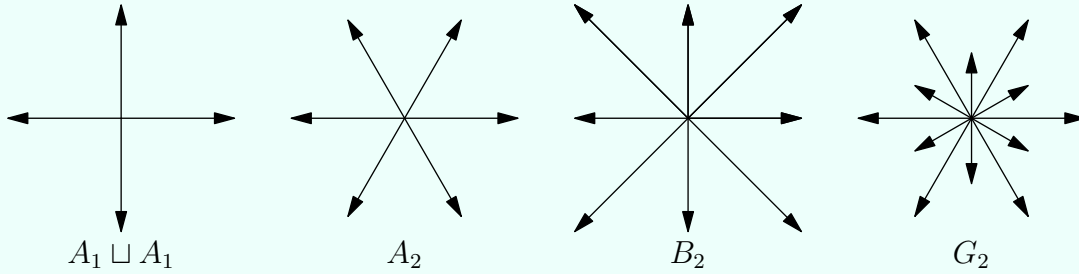
$$\alpha^\vee(\lambda) := \frac{2(\alpha, \lambda)}{(\alpha, \alpha)}.$$

Definition 71. Fix root systems $\Phi_1 \subseteq E_1$ and $\Phi_2 \subseteq E_2$. An *isomorphism* of root systems is a bijection $\varphi: \Phi_1 \rightarrow \Phi_2$ which extends to a linear map $E_1 \rightarrow E_2$ such that

$$\frac{2(\alpha, \beta)}{(\beta, \beta)} = \frac{2(\varphi(\alpha), \varphi(\beta))}{(\varphi(\beta), \varphi(\beta))}.$$

Remark 72. The linear map $\varphi: E_1 \rightarrow E_2$ does not have to preserve the inner product. For example, one could simply define E_2 to have twice the inner product of E_1 (and the same underlying vector space).

Example 73. Here are all the root systems of rank two, up to isomorphism, drawn in $\mathbb{R}^2$.



Definition 74 (irreducible). A root system Φ is *irreducible* if and only if it is not isomorphic to a root system of the form $\Phi_1 \sqcup \Phi_2$, where Φ_1 and Φ_2 are root systems.

Remark 75. It turns out that any root system is the disjoint union of irreducible root systems.

Definition 76 (Weyl group). Fix a root system $\Phi \subseteq E$. Then the *Weyl group* W of Φ is the subgroup of $GL(E)$ generated by $\{s_\alpha : \alpha \in \Phi\}$. The *sign* of $w \in W$, denoted $\det w$, is equal to the determinant of its action on E .

Example 77. The Weyl group of A_n is S_{n+1} .

1.6 Dynkin Diagrams

Definition 78 (polarization). Fix a root system $\Phi \subseteq E$. A *polarization* of Φ is a choice of $t \in E$ such that $(t, \alpha) \neq 0$ for all $\alpha \in \Phi$. The pair (Φ, t) is a *polarized root system*, though we will frequently omit the choice of t .

Definition 79 (positive). Fix a polarized root system $\Phi \subseteq E$. Then a root $\alpha \in \Phi$ is *positive* if and only if $(t, \alpha) > 0$. The set of positive roots is denoted Φ^+ . One similarly defines the set of negative roots Φ^- .

Example 80. The element $t := (1, 2, \dots, n)$ is a polarization of A_n . Then the positive roots are $\{e_i - e_j : i < j\}$.

Definition 81 (simple). Fix a polarized root system $\Phi \subseteq E$. Then a positive root $\alpha \in \Phi$ is *simple* if and only if it is not the sum of more than one positive root.

Remark 82. It turns out that the set of simple roots is a basis of E . Also, any positive root can be written as a sum of simple roots with nonnegative integral coefficients.

Example 83. The simple roots of A_n are $\{e_i - e_{i+1} : 1 \leq i < n\}$.

Lemma 84. Fix a root system $\Phi \subseteq E$. Then one can recover Φ from the subset $\Delta \subseteq \Phi$ of simple roots.

Sketch. Let W be the Weyl group of Φ . It turns out that the set $\{s_\alpha : \alpha \in \Delta\}$ of “simple reflections” generates W , and $W(\Delta) = \Phi$. ■

Definition 85 (Dynkin diagram). Fix a polarized root system $\Phi \subseteq E$, and enumerate the simple roots as $\{\alpha_1, \dots, \alpha_n\}$. Then the *Dynkin diagram* of Φ is the directed multigraph with the following data.

- The set of vertices of Φ is the index set $\{1, \dots, n\}$.
- Vertices i and j are connected by $a_{ij}a_{ji}$, with the edge directed from i to j if and only if $(\alpha_i, \alpha_i) > (\alpha_j, \alpha_j)$.

Remark 86. The Dynkin diagram D of a root system Φ is disconnected if and only if Φ is reducible, and each connected component of D corresponds to an irreducible component of Φ .

Example 87. The Dynkin diagram of A_n is the path with n vertices.

Lemma 88. Fix a root system $\Phi \subseteq E$. Then one can recover Φ from its Dynkin diagram.

Sketch. It is enough to recover the simple roots. Well, the Dynkin diagram remembers the number of simple roots and the numbers $n_{ij} := 2(\alpha_i, \alpha_j)/(\alpha_j, \alpha_j)$, which is enough to recover the root system up to isomorphism. ■

Theorem 89. The set of Dynkin diagrams of all possible irreducible root systems are as follows. (A graph labeled $\mathfrak{g}_n$ is understood to have n vertices.)

Name	Graph
A_n	$\bullet - \bullet - \dots - \bullet - \bullet$
B_n	$\bullet - \bullet - \dots - \bullet - \bullet$
C_n	$\bullet - \bullet - \dots - \bullet \Leftarrow \bullet$
D_n	$\bullet - \bullet - \dots - \bullet$ $\quad \quad \quad \diagup \bullet$ $\quad \quad \quad \diagdown \bullet$
E_4, E_5, E_6	$\bullet - \bullet - \bullet - \dots - \bullet$ $\quad \quad \quad \uparrow \bullet$
F_4	$\bullet - \bullet \Rightarrow \bullet - \bullet$
G_2	$\bullet \Rightarrow \bullet$

Sketch. There are two parts to this proof. First, one has to construct a root system giving each of these root systems. Second, one has to show that all Dynkin diagrams have this form. There is nothing too interesting about the first step. For the second step, the idea is to constrain the numbers $2(\alpha_i, \alpha_j)/(\alpha_i, \alpha_i)$ that can appear in a root system. For example, because $(-, -)$ is positive-definite, the matrix $[(\alpha_i, \alpha_j)]_{ij}$ is positive-definite. One can use this to rule out certain subgraphs from appearing in the Dynkin diagram, such as cycles. ■

Remark 90. By Remark 64 and Lemma 88, this classifies all simple complex Lie algebras.

1.7 Highest Weight Theory

Convention 91. Throughout this section, $\mathfrak{g}$ is a semisimple Lie algebra with a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. The polarized root system is $\Phi \subseteq \mathfrak{h}^*$, and its Weyl group is W . Representations need not be finite-dimensional.

Theorem 92. All finite-dimensional representations of $\mathfrak{g}$ are completely reducible.

Sketch. There are many proofs of this theorem. The main idea is to show that any short exact sequence

$$0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0$$

of representations of $\mathfrak{g}$ splits as a direct sum $V = V' \oplus V''$. The most algebraic proof proceeds by finding “Casimir” elements in the center of $U\mathfrak{g}$. By Lemma 12, they act by scalars on irreducible representations. One can force the splitting if granted a Casimir elements which acts on V' and V'' by different scalars. This turns out to be possible as long as V' and V'' are not isomorphic. The case where V' and V'' are isomorphic is handled directly. ■

Definition 93 (weight decomposition). A *weight* is an element of $\mathfrak{h}^*$. For a representation V of $\mathfrak{g}$, a *weight vector* is an eigenvector for $\mathfrak{h}$, and a *weight decomposition* is an eigenspace decomposition of $\mathfrak{h}$ acting on V . For $\lambda \in \mathfrak{h}^*$, we define $V[\lambda]$ to be the λ -eigenspace, meaning

$$V[\lambda] := \{v \in V : Xv = \lambda(X)v \text{ for all } X \in \mathfrak{h}\}.$$

The multiset of $\lambda \in \mathfrak{h}^*$ for which $V[\lambda] \neq 0$ are referred to as the *weights* of V , where multiplicities are by $\dim_{\mathbb{C}} V[\lambda]$.

Definition 94 (integral). A weight $\lambda \in \mathfrak{h}^*$ is *integral* if and only if $\alpha^\vee(\lambda) \in \mathbb{Z}$ for all $\alpha \in \Phi$. The subgroup of integral weights is denoted $P \subseteq \mathfrak{h}^*$. While we're here, we define the *root lattice* $Q \subseteq \mathfrak{h}^*$.

Remark 95. The definition of the root system implies that $Q \subseteq P$. The quotient P/Q is interesting.

Example 96. For $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{C})$, one has $P = \{(\lambda_1, \dots, \lambda_n) : \lambda_i - \lambda_j \in \mathbb{Z} \text{ for all } i, j\}$ and $Q = \{(\lambda_1, \dots, \lambda_n) : \sum_i \lambda_i = 0\}$. Thus, $P/Q \cong \mathbb{Z}/n\mathbb{Z}$.

Lemma 97. Any finite-dimensional representation V of $\mathfrak{g}$ admits a weight decomposition by integral weights. The multiset of weights of V is W -invariant.

Sketch. Let's start with the weight decomposition. Because $\mathfrak{h}$ is abelian and spanned by $\{h_\alpha\}_{\alpha \in \Phi}$, it is enough to show that h_α acts semisimply on V for each $\alpha \in \Phi$. For a given $\alpha \in \Phi$, this follows by the representation theory of $\mathfrak{sl}_2(\mathbb{C})$ upon restricting to $\mathfrak{sl}_2(\mathbb{C})_\alpha \subseteq \mathfrak{g}$. Integrality follows because h_α acts on $V[\lambda]$ by $\alpha^\vee(\lambda)$, which must be an integer. To check invariance under W , it is enough to check invariance under each reflection s_α for $\alpha \in \Phi$, which follows by the symmetry of the weights of the irreducible representations of $\mathfrak{sl}_2(\mathbb{C})_\alpha$. ■

Notation 98. Define $\mathfrak{n}_+$ to be

$$\bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha \subseteq \mathfrak{g}.$$

Define $\mathfrak{n}_-$ similarly. We may abbreviate $\mathfrak{n}_+$ to $\mathfrak{n}$. These are Lie subalgebras of $\mathfrak{g}$.

Definition 99 (singular). Fix a representation V of $\mathfrak{g}$. A vector $v \in V$ is *singular* if and only if v is a nonzero weight vector and $\mathfrak{n} \cdot v = 0$. The representation V is a *highest weight* representation of weight λ if and only if V is generated by a singular vector of weight λ .

Lemma 100. Fix an irreducible finite-dimensional representation V of $\mathfrak{g}$. Then V is a highest weight representation.

Sketch. By irreducibility, V is generated by any nonzero vector, so it is enough to exhibit a nonzero singular vector. Select the weight λ in V which is "maximal" in the sense that $\alpha + \lambda$ is not a weight in V for any $\alpha \in \Phi^+$. Then any nonzero $v \in V[\lambda]$ will do. ■

Definition 101 (Verma module). Fix a weight $\lambda \in \mathfrak{h}^*$. Then define the *Verma module* M_λ to be $U\mathfrak{g} \otimes_{\mathfrak{b}} \mathbb{C}_\lambda$, where $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}$. For brevity, we define $v_\lambda \in M_\lambda$ to be the vector $1 \otimes 1$.

Remark 102. By Theorem 16, the natural map $U\mathfrak{n}_- \rightarrow U\mathfrak{g}$ induces an isomorphism $U\mathfrak{n}_- \rightarrow M_\lambda$ of vector spaces, meaning that M_λ admits a basis by ordered monomials in $\mathfrak{n}_-$.

Lemma 103. Fix a representation V of highest weight λ . For any singular vector $v \in V[\lambda]$ which generates V , there is a unique surjection $M_\lambda \rightarrow V$ such that $v_\lambda \mapsto v$.

Sketch. There is a natural map $U\mathfrak{g} \rightarrow V$ sending $1 \mapsto v$. The properties of v imply that this natural map is surjective and descends to M_λ . ■

Lemma 104. Fix $\lambda \in \mathfrak{h}^*$. Then M_λ admits a unique maximal proper submodule.

Sketch. A submodule N of M_λ is proper if and only if $v_\lambda \notin N$, so the maximal submodule is simply the sum of all submodules not containing v_λ . ■

Notation 105. Fix a weight $\lambda \in \mathfrak{h}^*$. Then define L_λ to be the quotient of M_λ by its maximal proper submodule.

Remark 106. Assembling our results shows that any finite-dimensional irreducible representation V is of highest weight λ for some weight λ and thus must be isomorphic to L_λ .

Definition 107 (dominant). A weight $\lambda \in \mathfrak{h}^*$ is *dominant* if and only if $\alpha^\vee(\lambda) \geq 0$ for each $\alpha \in \Phi^+$. The set of dominant integral weights is P_+ .

Theorem 108 (Highest weight). For $\lambda \in \mathfrak{h}^*$, the representation L_λ is finite-dimensional if and only if λ is dominant and integral. Thus, finite-dimensional irreducible representations are in bijection with dominant integral weights.

Sketch. If L_λ is finite-dimensional, then for each $\alpha \in \Phi^+$, one knows that v_λ generates a finite-dimensional representation of $\mathfrak{sl}_2(\mathbb{C})_\alpha$ with $e_\alpha v_\lambda = 0$ and $h_\alpha v_\lambda = \alpha^\vee(\lambda)v_\lambda$. Comparing this with the known representations of $\mathfrak{sl}_2(\mathbb{C})$ forces $\alpha^\vee(\lambda)$ to be a nonnegative integer.

Conversely, suppose λ is dominant and integral, so we must check $\dim_{\mathbb{C}} L_\lambda < \infty$. This is long and difficult, so we will say very little. In short, for each simple reflection s , one can show that $M_{s\lambda}$ embeds into M_λ , and then one should check that the quotient of M by the sum of the $M_{s\lambda}$ s is finite-dimensional. ■

Remark 109. Though we have classified the finite-dimensional irreducible representations, this construction is highly non-explicit. It is possible but difficult to get rough information about these representations; see Theorem 119 and Corollary 121.

1.8 Character Theory

Convention 110. Convention 91 continues to apply.

Definition 111 (character). Fix a representation V of $\mathfrak{g}$. Then the *character* of V is the formal power series

$$\text{ch } V := \sum_{\lambda \in \mathfrak{h}^*} \dim_{\mathbb{C}} V[\lambda] \cdot e^{\lambda}.$$

Here, e^{λ} is a formal symbol satisfying the usual exponent rule $e^{\lambda+\mu} = e^{\lambda}e^{\mu}$ for $\lambda, \mu \in \mathfrak{h}^*$.

Example 112. Let $\mathbb{C}$ be the trivial representation. Then $\text{ch } \mathbb{C} = 1$.

Example 113. Let $\mathbb{C}^2$ be the standard representation of $\mathfrak{sl}_2(\mathbb{C})$. Then $\text{ch } \mathbb{C}^2 = e^{-\alpha/2} + e^{\alpha/2}$, where $\alpha \in \mathfrak{h}$ is the unique positive root. More generally,

$$\text{ch Sym } \mathbb{C}^2 = e^{-n\alpha/2} + e^{-n\alpha/2+1} + \dots + e^{n\alpha/2-1} + e^{n\alpha/2}.$$

Example 114. For any representation V , one has $\text{ch } V^{\vee} = \overline{\text{ch } V}$, where the overline means $\overline{e^{\lambda}} = e^{-\lambda}$.

Example 115. For representations V and W , one has $\text{ch}(V \oplus W) = \text{ch } V + \text{ch } W$. More generally, given a short exact sequence

$$0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0$$

of representations admitting weight decompositions, one has $\text{ch } V = \text{ch } V' + \text{ch } V''$.

Example 116. For representations V and W , one has $\text{ch}(V \otimes W) = \text{ch } V \cdot \text{ch } W$.

Example 117. Fix $\lambda \in \mathfrak{h}^*$. By expanding out the basis of M_{λ} , one finds that

$$\text{ch } M_{\lambda} = e^{\lambda} \prod_{\alpha \in \Phi^+} \frac{1}{1 - e^{-\alpha}}.$$

Notation 118. Define $\rho := \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha$.

Theorem 119 (Weyl character formula). Fix a dominant integral weight λ . Then

$$\text{ch } L_{\lambda} = \sum_{w \in W} \det(w) e^{w(\lambda+\rho)} \bigg/ \prod_{\alpha \in \Phi^+} (e^{\alpha/2} - e^{-\alpha/2}).$$

Sketch. This is another long and difficult theorem. One approach is to build an exact sequence of the form

$$0 \rightarrow ? \rightarrow ? \rightarrow \dots \rightarrow ? \rightarrow L_{\lambda} \rightarrow 0,$$

where each $?$ is a sum of Verma modules. For example, there is a natural surjection $M_{\lambda} \rightarrow L_{\lambda}$, and the kernel admits a natural surjection of the form $\bigoplus_i M_{s_i \lambda}$. Once one fills in each $?$, one can apply Example 117. ■

Example 120 (Weyl demoninator formula). When $\lambda = 0$, we find $L_0 = \mathbb{C}$, so $\text{ch } L_0 = 1$. Thus,

$$\sum_{w \in W} \det(w) e^{w\rho} = \prod_{\alpha \in \Phi^+} (e^{\alpha/2} - e^{-\alpha/2}).$$

Corollary 121 (Weyl dimension formula). Fix a dominant integral weight λ . Then

$$\dim L_\lambda = \prod_{\alpha \in \Phi^+} \frac{(\lambda + \rho, \alpha)}{(\rho, \alpha)}.$$

Sketch. Fix a formal variable q , and we will compute

$$\dim_q L_\lambda := \sum_{\mu \in \mathfrak{h}^*} \dim_{\mathbb{C}} L_\lambda[\mu] \cdot q^{2(\rho, \lambda)},$$

from which the dimension can be computed by sending $q \rightarrow 1$. There is a homomorphism $\mathbb{C}[e^P] \rightarrow \mathbb{C}[q, 1/q]$ given by sending e^λ to $q^{2(\lambda, \rho)}$, and it sends $\text{ch } L_\lambda$ to $\dim_q L_\lambda$. One now computes $\dim_q L_\lambda$ by using Theorem 119: note

$$\dim_q L_\lambda = \sum_{w \in W} \det(w) q^{(w(\lambda + \rho), \rho)} \Bigg/ \prod_{\alpha \in \Phi^+} (q^{(\rho, \alpha)} - q^{-(\rho, \alpha)}).$$

The numerator can be simplified with Example 120, and then we may take $q \rightarrow 1$. ■

2 Lie Groups

We now allow for vector spaces to be real or complex. Notably, in contrast to the previous section, our Lie algebras will mostly be real; many of the results of Section 1 continue to apply, but we will not take care to distinguish them. (As a rule of thumb, sections 1.1 to 1.3 work if $\mathfrak{g}$ is real.) Let's overview where we're going.

- Sections 2.1 and 2.2 provide the basic definitions and constructions with Lie groups.
- Sections 2.3 and 2.4 constructs the Lie algebra associated to a Lie group and provides its basic properties.
- Section 2.5 discusses how to recover information about a Lie group from its Lie algebra.
- Section 2.6 discusses the representation theory of compact Lie groups. It is optional and harder.

2.1 Starting Out

Definition 122 (Lie group). A *Lie group* is a manifold G equipped with a distinguished element $1 \in G$, a smooth multiplication map $m: G \times G \rightarrow G$, and a smooth inversion map $i: G \rightarrow G$. These maps are required to make G into a group in the natural way. A homomorphism of Lie groups is a smooth map preserving the group structure.

Remark 123. One can imagine weakening the hypotheses of a Lie group. As long as G is a manifold, as soon as the group operations are continuous, one can prove that they are smooth and even real analytic.

Example 124. Let $\mathrm{GL}_n(\mathbb{R})$ denote the (open) subset of $n \times n$ invertible matrices. It is an open submanifold and in fact a Lie group.

Example 125. For any bilinear form B on $\mathbb{R}^n$, the orthogonal group $O(B)$ preserving the bilinear form is a closed submanifold of $\mathrm{GL}_n(\mathbb{R})$. For example, if B is positive-definite, we denote this group by $O_n(\mathbb{R})$. If B is symplectic (i.e., non-degenerate and alternating), we denote this group by $\mathrm{Sp}_{2n}(\mathbb{R})$.

Example 126. Similarly, the group $U(n)$ of $n \times n$ unitary matrices forms a Lie group.

Notation 127. Fix a Lie group G . For any $g \in G$, let $L_g: G \rightarrow G$ denote left translation by g . Define $R_g: G \rightarrow G$ by $R_g: h \mapsto hg^{-1}$. Lastly, for $g \in G$, let $\mathrm{Ad}_g: G \rightarrow G$ denote the *adjoint action* $\mathrm{Ad}_g: h \mapsto ghg^{-1}$.

Definition 128 (representation). Fix a Lie group G . Then a *representation* of G is a complex vector space V along with a homomorphism $G \rightarrow \mathrm{GL}(V)$.

Example 129 (adjoint). For each $g \in G$, the adjoint action $\mathrm{Ad}_g: G \rightarrow G$ fixes $1 \in G$ and thus induces a linear map $d\mathrm{Ad}_g: T_1G \rightarrow T_1G$. This induces the *adjoint representation* $\mathrm{Ad}_\bullet: G \rightarrow \mathrm{GL}(T_1G)$.

Example 130. As usual, one can take sums, tensor products, and duals of representations. For example, if V and W are representations of G , then G acts on the tensor product $V \otimes W$ by

$$g(v \otimes w) := gv \otimes gw.$$

Remark 131. As in Remark 5, it is an interesting question if all Lie groups G can be embedded into $\mathrm{GL}_n(\mathbb{R})$ (or $\mathrm{GL}_n(\mathbb{C})$) for all n . The answer turns out to be no! For example, the universal cover $\tilde{G}$ of $\mathrm{SL}_2(\mathbb{R})$ turns out to be a Lie group, and it admits no embedding into $\mathrm{GL}_n(\mathbb{C})$ for any n . One way to show this is to show that any representation $\tilde{G} \rightarrow \mathrm{GL}_n(\mathbb{C})$ descends to a representation of $\mathrm{SL}_2(\mathbb{R})$ (using an elaboration of Remark 169) and thus has nontrivial kernel.

2.2 Subgroups and Quotients

Definition 132 (closed Lie subgroup). Fix a Lie group G . Then a *closed Lie subgroup* is a submanifold $H \subseteq G$ which is also a Lie subgroup.

Remark 133. It turns out that any closed Lie subgroup H of G is a closed submanifold. In short, this means $\overline{H}$ is still a subgroup, and any coset of $\overline{H}/H$ is open and dense in $\overline{H}$, so $H = \overline{H}$. Once H is closed, one can show that H is a closed submanifold of G by explicitly constructing a slice chart of H around $1 \in G$. (This is harder than it sounds and uses the so-called exponential map.)

Example 134 (kernel). Fix a homomorphism $\varphi: G \rightarrow H$ of Lie groups. Then the kernel K of φ is a closed Lie subgroup. In particular, it is a manifold because φ has constant rank.

Example 135 (center). Fix a connected Lie group G . Then the center $Z(G) \subseteq G$ is a closed Lie subgroup. Indeed, $Z(G)$ is the kernel of the adjoint representation $\mathrm{Ad}_\bullet: G \rightarrow \mathrm{GL}(T_1 G)$.

Example 136 (identity component). Fix a Lie group G . Then let G° denote the *identity component*, which is the open (and closed) submanifold of G consisting of the connected component of the identity. It is a closed Lie subgroup.

Remark 137. It turns out that $G^\circ \subseteq G$ is a normal subgroup. This follows from continuity of multiplication (and the fact that $1 \cdot 1 = 1$).

Definition 138 (component group). Fix a Lie group G . Then let $\pi_0 G$ denote the *component group*, which is the discrete set indexed by the connected components of G .

Remark 139. For any Lie group G , there is a short exact sequence of Lie groups

$$1 \rightarrow G^\circ \rightarrow G \rightarrow \pi_0 G \rightarrow 1.$$

Thus, one can frequently reduce the study of general Lie groups to the study of connected ones.

Lemma 140. Fix a connected Lie group G . Then the universal cover $\tilde{G}$ is also a manifold, and it admits a unique structure of a Lie group for which the canonical projection $\pi: \tilde{G} \rightarrow G$ is a homomorphism.

Sketch. Note $\tilde{G}$ is a manifold because it is locally Euclidean and second countable; second countability holds because $\pi_1(G)$ is finitely generated. One constructs the group structure on $\tilde{G}$ by “path-lifting” the group structure on G . ■

Remark 141. Fix a connected Lie group G . Let K be the kernel of $\pi: \tilde{G} \rightarrow G$. Then there is a short exact sequence

$$1 \rightarrow Z \rightarrow \tilde{G} \xrightarrow{\pi} G \rightarrow 1.$$

- For dimensions reasons, Z is discrete.
- A path-lifting argument shows that $Z \cong \pi_1(G)$.
- By studying the conjugation action of $\tilde{G}$ on Z , one sees that $Z \subseteq Z(\tilde{G})$.

Thus, one can frequently reduce the study of general connected Lie groups to the study of simply connected ones.

Proposition 142. Fix a Lie group G and a closed Lie subgroup $H \subseteq G$. Then G/H admits a structure of a manifold of dimension $\dim G - \dim H$ for which the canonical map $\pi: G \rightarrow G/H$ is a fiber bundle with fiber H .

Sketch. By translation, it is enough to exhibit a chart in an open neighborhood of $1 \in G/H$. One does this by choosing an open neighborhood U of $1 \in G$ small enough so that one can find a closed submanifold $M \subseteq G$ where

$$dm: T_1 H \oplus T_1 M \rightarrow T_1 G$$

is an isomorphism. Then $M \rightarrow G/H$ is a homeomorphism and provides the needed manifold structure around $1 \in G/H$. ■

2.3 The Exponential

Definition 143 (Lie algebra). Fix a Lie group G . Then we define the *Lie algebra* $\text{Lie } G$ of G to be the tangent space at the identity. It is a real vector space.

Remark 144. By translation, $\text{Lie } G$ is isomorphic to the vector space of right-invariant vector fields. One can use this to define the Lie bracket on $\text{Lie } G$ by using the Lie bracket of vector fields. We will take an approach using less differential geometry, but this is useful to know.

Remark 145. A choice of basis of $\text{Lie } G$ induces a parallelization on G by left-invariant vector fields. For example, this implies that G is orientable, so one can do integration on G .

Example 146. Let $\varphi: G \rightarrow H$ be a homomorphism, and let K be the kernel. Then φ has constant rank, so

$$\text{Lie } K = \{X \in \text{Lie } G : d\varphi_1(X) = 0\}.$$

For example, if G is connected, then $\text{Lie } Z(G) = \mathfrak{z}(\mathfrak{g})$.

Definition 147 (exponential). Fix a Lie group G with Lie algebra $\mathfrak{g}$. Then the *exponential* $\exp_G: \mathfrak{g} \rightarrow G$ is defined as follows: for $X \in \mathfrak{g}$, let $\gamma_X: \mathbb{R} \rightarrow G$ be the unique Lie group homomorphism such that $\gamma'_X(0) = X$. Then $\exp_G(X) := \gamma_X(1)$. We will abbreviate $\exp_G$ to $\exp$.

Remark 148. The requirement that γ_X be a homomorphism means that γ_X is a path following the flow of the left-invariant vector field determined by X . This explains its existence and uniqueness. One can check that $\exp$ is a local diffeomorphism.

Example 149. For the group $G = \mathrm{GL}_n(\mathbb{R})$, the exponential $\exp: M_n(\mathbb{R}) \rightarrow G$ is the matrix exponential.

Lemma 150. Fix a homomorphism $\varphi: G \rightarrow H$ of Lie groups. For any $X \in \mathrm{Lie} G$, one has

$$\varphi(\exp_G(X)) = \exp_H(d\varphi_1(X)).$$

Sketch. This follows from the uniqueness of the construction γ_X for $X \in \mathrm{Lie} G$. ■

Example 151. For any representation $G \rightarrow \mathrm{GL}_n(\mathbb{R})$, we see that the exponential $\mathfrak{g} \rightarrow G$ can be realized as a restriction of the matrix exponential $\mathfrak{gl}_n(\mathbb{R}) \rightarrow \mathrm{GL}_n(\mathbb{R})$.

Example 152. Let's compute $\mathfrak{o}_n(\mathbb{R}) := \mathrm{Lie} O_n(\mathbb{R})$. (Example 146 does not apply.) By definition, $O_n(\mathbb{R}) \subseteq \mathrm{GL}_n(\mathbb{R})$ is the stabilizer of the usual inner product B on $\mathbb{R}^n$; one can view B as the identity matrix in $\mathbb{R}^{n \times n}$. The action map of $\mathrm{GL}_n(\mathbb{R})$ on the $\mathbb{R}^{n \times n}$ has constant rank, so we may calculate

$$\mathfrak{o}_n(\mathbb{R}) = \{X \in \mathfrak{gl}_n(\mathbb{R}) : X \cdot B = 0\}.$$

Here, this action is by $X \cdot B = \frac{d}{dt} \exp(tX)^T B \exp(tX) \big|_{t=0}$. An application of the product rule yields $\mathfrak{o}_n(\mathbb{R}) = \{X \in \mathfrak{gl}_n(\mathbb{R}) : X^T B + BX = 0\}$, so $\mathfrak{o}_n(\mathbb{R})$ consists of the alternating matrices. A similar discussion holds for other bilinear forms B .

Example 153 (adjoint group). Fix a Lie algebra $\mathfrak{g}$. Then define $\mathrm{Aut}(\mathfrak{g}) \subseteq \mathrm{GL}(\mathfrak{g})$ to be the automorphism group of $\mathfrak{g}$. Namely, $g \in \mathrm{Aut}(\mathfrak{g})$ if and only if g preserves the Lie bracket. An argument similar to Example 152 shows that

$$\mathrm{Lie} \mathrm{Aut}(\mathfrak{g}) = \{D \in \mathfrak{gl}(\mathfrak{g}) : [DX, Y] + [X, DY] = D[X, Y]\}.$$

The right-hand space is known as the Lie algebra of *derivations* of $\mathfrak{g}$ and is denoted $\mathrm{Der}(\mathfrak{g})$. If $\mathfrak{g}$ is semisimple, then the adjoint map $\mathrm{ad}_\bullet: \mathfrak{g} \rightarrow \mathrm{Der}(\mathfrak{g})$ is an isomorphism. (The idea is to decompose $\mathrm{Der}(\mathfrak{g})$ as a representation of $\mathfrak{g}$ as $\mathfrak{g} \oplus I$ and then show that any $D \in I$ acts on $\mathfrak{g}$ by zero.) Thus, when $\mathfrak{g}$ is semisimple, we define the *adjoint group* G_{ad} to be $\mathrm{Aut}(\mathfrak{g})^\circ$, which has $\mathrm{Lie} G_{\mathrm{ad}} \cong \mathfrak{g}$.

Remark 154. Suppose G is a connected Lie group with semisimple Lie algebra $\mathfrak{g}$. Then the adjoint representation $\mathrm{Ad}_\bullet: G \rightarrow \mathrm{GL}(\mathfrak{g})$ restricts to $G \rightarrow G_{\mathrm{ad}}$, which then has finite kernel, so $G \rightarrow G_{\mathrm{ad}}$ is a covering space. It follows that G_{ad} is the unique connected Lie group with trivial center and Lie algebra $\mathfrak{g}$.

2.4 The Commutator

Definition 155 (commutator). Fix a Lie group G with Lie algebra $\mathfrak{g}$. Then we define the *Lie bracket* $[-, -]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ as the unique bilinear map fitting into the Taylor expansion satisfying

$$\exp(X)\exp(Y) = \exp\left(X + Y + \frac{1}{2}[X, Y] + \cdots\right)$$

for small $X, Y \in \mathfrak{g}$. This definition makes sense because $\exp$ is a local diffeomorphism. We may write $[X, Y]$ as $\mathrm{ad}_X(Y)$.

Remark 156. By plugging in $X = Y$, one finds that $[-, -]$ is alternating.

Example 157. By explicit calculation, for $X, Y \in \text{Lie } \text{GL}_n(\mathbb{R})$, we have $[X, Y] = XY - YX$. This supports the intuition that $[-, -]$ measures how “noncommutative” G is.

Lemma 158. Fix a homomorphism $\varphi: G \rightarrow H$ of Lie groups. For any $X, Y \in \text{Lie } G$, one has

$$[d\varphi_1(X), d\varphi_1(Y)] = d\varphi_1([X, Y]).$$

Sketch. Use Lemma 150. ■

Example 159. For any representation $G \rightarrow \text{GL}_n(\mathbb{R})$, we see that the Lie bracket on $\text{Lie } G$ can be viewed as the restriction of the bracket on $\mathfrak{gl}_n(\mathbb{R})$.

Proposition 160. Fix a Lie group G with Lie algebra $\mathfrak{g}$. Then the Lie bracket $[-, -]$ makes $\mathfrak{g}$ into a Lie algebra.

Sketch. It only remains to check the Jacobi identity. It is enough to show that $\text{ad}_{[X, Y]} Z = \text{ad}_X \text{ad}_Y Z - \text{ad}_Y \text{ad}_X Z$. Well, the adjoint representation $\text{Ad}_\bullet: G \rightarrow \text{GL}(\mathfrak{g})$ has

$$\text{Ad}_g(Y) = \left. \frac{d}{dt} g \exp(tY) g^{-1} \right|_{t=0}.$$

Differentiating with respect to g gives $(d \text{Ad}_1)_X(Y) = \text{ad}_X(Y)$. We conclude that $\text{ad}_{[X, Y]} = [\text{ad}_X, \text{ad}_Y]$, where the second bracket takes place in $\mathfrak{gl}(\mathfrak{g})$. ■

Remark 161. By Proposition 160, we see Lie is a functor from Lie groups to Lie algebras.

Theorem 162 (Baker–Campbell–Hausdorff formula). Fix a Lie group G with Lie algebra $\mathfrak{g}$. Then there are “Lie polynomials” (i.e., iterated Lie brackets) $\{\mu_n\}_{n \geq 1}$, where $\deg \mu_n = n$, satisfying the equality

$$\exp(X) \exp(Y) = \exp \left(x + y + \sum_{n=1}^{\infty} \mu_n(x, y) \right)$$

for small $X, Y \in \mathfrak{g}$. In fact, the expressions $\{\mu_n\}_n$ are independent of G and $\mathfrak{g}$.

Sketch. This is another long and difficult theorem. Roughly speaking, the approach is to formally expand the power series of $\log(\exp(X) \exp(Y))$ and group terms together with Lie brackets. For example, one can reduce to the case $\mathfrak{gl}_n(\mathbb{R})$ via Ado’s theorem (Remark 5) and then use the power series expansions for the matrix logarithm and exponential. ■

2.5 The Fundamental Theorems

Proposition 163. Fix Lie groups G and H with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. If G is connected, then

$$\mathrm{Hom}_{\mathrm{LieGrp}}(G, H) \rightarrow \mathrm{Hom}_{\mathrm{LieAlg}}(\mathfrak{g}, \mathfrak{h})$$

is injective.

Sketch. Any homomorphism $\varphi: G \rightarrow H$ has constant rank. If $d\varphi_1 = 0$, then $\ker \varphi$ has equal dimension to G , so $\ker \varphi = G$ because G is connected, so $\varphi = 1$. ■

Example 164. The connectivity of G is a necessary hypothesis. For example, for any Lie group G , the natural map $G \rightarrow \pi_0 G$ vanishes on the Lie algebra but is nontrivial whenever G is not connected.

Remark 165. It is an interesting question how much information about Lie groups can be recovered from the Lie algebra. Proposition 163 tells us that the functor Lie is faithful on connected Lie groups. We will soon see (in Corollary 171) that the functor is an equivalence on simply connected Lie groups.

Theorem 166 (Lie's first). Fix a Lie group G . Then the map $H \mapsto \mathrm{Lie} H$ is a bijection between connected closed Lie subgroups $H \subseteq G$ and Lie subalgebras $\mathfrak{h} \subseteq \mathfrak{g}$.

Sketch. Choose a Lie subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Uniqueness of H is not so hard: if H and H' are connected and have the same Lie algebra, then $H/(H \cap H')$ is connected and discrete and hence trivial, so $H = H \cap H'$; similarly, $H' = H \cap H'$.

Existence of H is tricky for reasons related to differential geometry. The idea is to extend $\mathfrak{h}$ to a family of right-invariant vector fields. One can use the exponential to define H in a neighborhood of $1 \in G$, and then one uses the integrability of the vector fields to translate this definition of H into a genuine subgroup. ■

Theorem 167 (Lie's second). Fix Lie groups G and H . If G is connected and simply connected, then the natural map

$$\mathrm{Hom}_{\mathrm{LieGrp}}(G, H) \rightarrow \mathrm{Hom}_{\mathrm{LieAlg}}(\mathfrak{g}, \mathfrak{h})$$

is bijective.

Sketch. Proposition 163 gives injectivity. For surjectivity, choose some $d\varphi: \mathfrak{g} \rightarrow \mathfrak{h}$ to lift. Let $\Gamma(d\varphi) \subseteq \mathfrak{g} \oplus \mathfrak{h}$ be the graph of $d\varphi$. It is a Lie subalgebra, so Theorem 166 provides a connected Lie subgroup $\Gamma \subseteq G \times H$. We now define $\varphi: G \rightarrow H$ as the composite

$$G \leftarrow \Gamma \rightarrow H.$$

The left map is well-defined because $\pi: \Gamma \rightarrow G$ is an isomorphism: $d\pi: \Gamma(d\varphi) \rightarrow \mathfrak{g}$ is an isomorphism, so $\ker \pi$ is discrete, so π is a covering space, so π is a homeomorphism because G is simply connected. ■

Example 168. Fix a connected and simply connected Lie group G with Lie algebra $\mathfrak{g}$. By Theorem 167, the category of representations of G is equivalent to the category of representations of $\mathfrak{g}$.

Remark 169. If G is a connected Lie group, then G can be written as $\tilde{G}/Z$ where $\tilde{G}$ is the universal cover, and $Z \subseteq \tilde{G}$ is central and discrete. A representation of $\mathrm{Lie} G$ always lifts to a representation of $\tilde{G}$, but it only descends to G if Z acts trivially.

Theorem 170 (Lie's third). Fix a (finite-dimensional) Lie algebra $\mathfrak{g}$. Then there is a Lie group G with $\text{Lie } G \cong \mathfrak{g}$.

Sketch. One approach is to use Ado's theorem (Remark 5). Alternatively, one can follow the given sketch of Ado's theorem: handle semisimple $\mathfrak{g}$ via the adjoint representation (see Example 153), handle solvable $\mathfrak{g}$ via an inductive argument, and combine these two to achieve the general case via Theorem 45. ■

Corollary 171. The category of simply connected Lie groups is equivalent to the category of Lie algebras.

Sketch. Combine Theorems 166, 167 and 170. ■

2.6 ★ Compact Lie Groups



Warning 172. This section is optional, but it is my favorite topic. We will provide more details than usual.

Definition 173 (Haar measure). Fix a Lie group G . Then the line bundle $\wedge^{\dim G} \Omega_G$ of top-differential forms admits a unique left-invariant global section ω_G , up to scalar. The induced left-invariant Radon measure μ_G is known as the (left) Haar measure.

Example 174. In general, μ_G does not have to be right-invariant. For example, this happens if G is the subgroup of upper-triangular matrices in $\text{GL}_n(\mathbb{R})$.

Lemma 175. Let G be a compact Lie group. Then μ_G is also right-invariant.

Sketch. For any $g \in G$, the measure $U \mapsto \mu_G(Ug)$ is still left-invariant and thus equals μ_G times a positive scalar $\delta_G(g_0)$. One can show that δ_G is a continuous homomorphism $G \rightarrow \mathbb{R}^+$, which must be trivial because G is compact. ■

Proposition 176. Fix a compact Lie group G . Then any representation of G is completely reducible.

Sketch. Fix a representation V of G . It is enough to show that any invariant subspace $W \subseteq V$ admits an invariant complement W' for which $V = W \oplus W'$. To this end, it is enough to exhibit an invariant positive-definite Hermitian form on V (and then take orthogonal complements). Well, start with any positive-definite Hermitian form $\langle -, - \rangle_0$ on V . Then

$$\langle v, w \rangle := \int_G \langle gv, gw \rangle_0 d\mu_G(g)$$

is an invariant positive-definite Hermitian form. ■

Corollary 177. Fix a compact Lie group G . Then $\text{Lie } G$ is reductive.

Sketch. Set $\mathfrak{g} := (\text{Lie } G)_{\mathbb{C}}$. It is enough to show $\mathfrak{g}$ is reductive. Well, use Proposition 176 to decompose $\mathfrak{g}$ as a representation of G° . The irreducible components are either abelian or simple, so $\mathfrak{g}$ is reductive. ■

Proposition 178. Fix a complex semisimple Lie algebra $\mathfrak{g}$. Then there is a compact Lie group G_{ad}^c with $(\text{Lie } G)_{\mathbb{C}} \cong \mathfrak{g}$.

Sketch. Start with the complex manifold G_{ad} . Then define the real Lie subalgebra $\mathfrak{g}^c \subseteq \mathfrak{g}$ to be the antilinear fixed points of the “Cartan involution” $\tau: \mathfrak{g} \rightarrow \mathfrak{g}$ defined by

$$\begin{cases} \tau(e_i) = -f_i, \\ \tau(f_i) = -e_i, \\ \tau(h_i) = -h_i. \end{cases}$$

Here, $i \in \{1, 2, \dots, n\}$ is an index set for the simple roots of $\mathfrak{g}$. (Technically, $\mathfrak{g}^c = \{X \in \mathfrak{g} : \tau(X) = \overline{X}\}$.) The closed Lie subgroup $G_{\text{ad}}^c \subseteq G_{\text{ad}}$ cut out by $\mathfrak{g}^c$ will do. For example, G_{ad}^c is compact because the Killing form is negative-definite on $\mathfrak{g}^c$. ■

Remark 179. When Proposition 178 is combined with Proposition 176, this gives another proof of Theorem 92. A similar argument works for reductive Lie algebras.

Definition 180 (matrix coefficient). Fix a representation V of a Lie group G . For any $v \in V$ and $\ell \in V^\vee$, we define the *matrix coefficient* to be the map $g \mapsto \ell(gv)$.

Theorem 181 (Peter–Weyl). Fix a compact Lie group G , and let $\text{IrRep}(G)$ denote the collection of finite-dimensional irreducible representations of G . Then the matrix coefficient map

$$\bigoplus_{V \in \text{IrRep}(G)} (V \otimes V^\vee) \rightarrow L^2(G)$$

has dense image.

Sketch. This is a long and difficult theorem. The map is G -invariant, where G acts on V on the left and on $L^2(G)$ by right translation. Injectivity of this map is easier: by complete reducibility, it is enough to show that the matrix coefficient map defines an isomorphism

$$V^\vee \cong \text{Hom}_G(V, L^2(G)).$$

Roughly speaking, the inverse map sends $f: V \rightarrow L^2(G)$ to the linear functional $v \mapsto f(v)(1)$.

Surjectivity is more difficult. It is not too hard to show that the image $L^2(G)^{\text{fin}}$ of the matrix coefficient map consists of those $f \in L^2(G)$ for which

$$\dim_{\mathbb{C}} \text{span}\{gf : g \in G\} < \infty,$$

so it remains to show that $L^2(G)^{\text{fin}}$ is dense. The idea is to use the Spectral theorem for compact self-adjoint operators to decompose $L^2(G)$ into eigenspaces for the convolution operators $R\varphi: L^2(G) \rightarrow L^2(G)$ defined by

$$R\varphi(f)(g) := \int_G \varphi(x) f(xg) dx,$$

where $\varphi \in C(G)$. It turns out to be enough to choose $\varphi \in C(G)$ from an approximate identity of G . ■

Corollary 182. Fix a compact Lie group G . Then G admits a finite-dimensional faithful representation. In other words, there is an injective homomorphism $G \rightarrow \mathrm{GL}_n(\mathbb{C})$ for some n .

Sketch. The action $G \rightarrow \mathrm{GL}(L^2(G))$ is faithful. Using the compactness of G , one can then show that the action of G on some finite-dimensional subrepresentation of

$$L^2(G) = \bigoplus_{V \in \mathrm{IrRep}(G)} (V \otimes V^\vee)$$

must be faithful. (We used Theorem 181 here.) ■

Corollary 183. Fix a compact Lie group G , and choose a faithful representation $G \rightarrow \mathrm{GL}(V)$. Then any representation W of G is a summand of a sum of tensor powers of V and $V^\vee$.

Sketch. All finite-dimensional irreducible representations of G are found in $L^2(G)^{\mathrm{fin}}$, so it is enough to show that the algebra $A(V) \subseteq L^2(G)$ generated by the matrix coefficients of V and $V^\vee$ equals $L^2(G)^{\mathrm{fin}}$. In fact, it is enough to show that $A(V)$ is dense in $C(G)$. This is then an application of the Stone–Weierstrass theorem: $A(V)$ separates points of G because V is faithful, and $A(V)$ is stable under complex conjugation because the conjugate of a matrix coefficient of V is found as a matrix coefficient of $V^\vee$. ■

Proposition 184. Fix a compact Lie group G . Then $\mathrm{rank} \pi_1(G) = \dim Z(G)$.

Sketch. One can apply the homotopy long exact sequence to the short exact sequence

$$1 \rightarrow Z(G) \rightarrow G \rightarrow G/Z(G) \rightarrow 1$$

to reduce to the case where $Z(G)$ is trivial. (To compute $\pi_1(Z(G))$, one needs the fact that a connected abelian compact Lie group is a product of S^1 s.) We now assume $Z(G) = 1$, and we will show $\pi_1(G)$ is finite. Note $\mathfrak{g} := (\mathrm{Lie} G)_{\mathbb{C}}$ is semisimple.

Let G_{sc} be the universal cover of G . The kernel $Z \subseteq G_{\mathrm{sc}}$ of the projection is discrete and central. We will show Z is finite by showing it is isomorphic to $\mathrm{Hom}(P/Q, \mathbb{C}^\times)$. There is a map $\chi_\bullet: P_+ \rightarrow \mathrm{Hom}(Z, \mathbb{C}^\times)$ sending $\lambda \in P_+$ to the central character χ_λ of Z acting on L_λ . Because $L_{\lambda+\mu} \subseteq L_\lambda \otimes L_\mu$ for $\lambda, \mu \in P_+$, we see $\chi_\bullet$ extends to a character $P \rightarrow \mathrm{Hom}(Z, \mathbb{C}^\times)$.

The proof will be complete once we show that $\chi_\bullet$ descends to an isomorphism $P/Q \rightarrow \mathrm{Hom}(Z, \mathbb{C}^\times)$.

- One shows $Q \subseteq \ker \chi_\bullet$ by showing that $L_\lambda \subseteq L_\lambda \otimes L_\alpha$, where $\alpha \in \Phi$ and λ is large enough.
- One shows $\ker \chi_\bullet \subseteq Q$ by using Corollary 183 and the fact that $\mathfrak{g}$ is a faithful representation of G .
- One shows $\chi_\bullet$ is surjective with Theorem 181. Actually, one uses the density of $L^2(G)^{\mathrm{alg}}$ in $C(G)$. ■

Example 185. Fix a complex semisimple Lie algebra $\mathfrak{g}$. Then there are only finitely many compact Lie groups G with $(\mathrm{Lie} G)_{\mathbb{C}} \cong \mathfrak{g}$. Indeed, one can show that any such compact Lie group G covers G_{ad}^c , which has finite fundamental group by Proposition 184.

A Type A_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{sl}(n+1)$.
- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_{n+1}) : a_1 + \dots + a_{n+1} = 0\}$.
- $\mathfrak{h}^*$ is spanned by the projections $e_i : \text{diag}(a_1, \dots, a_{n+1}) \mapsto a_i$, where $e_1 + \dots + e_{n+1} = 0$.
- Killing form $K(X, Y) = 2(n+1) \text{tr } XY$.

- Root system

root	root space	coroot
$e_i - e_j$	E_{ij}	$E_{ii} - E_{jj}$

- Lattices

- Weight lattice $P = \{(\lambda_1, \dots, \lambda_{n+1}) \in \mathbb{R}^{n+1} / (1, \dots, 1)\mathbb{R} : \lambda_i - \lambda_{i+1} \in \mathbb{Z} \text{ for all } i\}$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n\}$.
- Root lattice $Q = \{(\lambda_1, \dots, \lambda_{n+1}) \in \mathbb{Z}^{n+1} : \lambda_1 + \dots + \lambda_{n+1} = 0\}$.
- $P/Q \cong \mathbb{Z}/(n+1)\mathbb{Z}$ by $(\lambda_1, \dots, \lambda_{n+1}) \mapsto \sum_i \lambda_i$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\}$.
- The simple roots are $\{e_i - e_{i+1} : i\}$. Set $\alpha_i := e_i - e_{i+1}$.
- Dynkin diagram:

$$\alpha_1 \text{ --- } \alpha_2 \text{ --- } \dots \text{ --- } \alpha_n$$

- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones.
- Maximal root $\theta = e_{n+1} - e_n$.

- Combinatorics

- Weyl group is $W = S_{n+1}$ acting by permutations of coordinates.
- The simple reflection s_i is $(i, i+1)$.
- $w_0 = (n+1, n, \dots, 2, 1)$.

- Representation theory

- $\rho = (n/2, n/2 - 1, \dots, 1 - n/2, -n/2)$ and $\rho^\vee = (n/2, n/2 - 1, \dots, 1 - n/2, -n/2)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_i : i\}$.
- Fundamental representations are $L_{\omega_i} = \wedge^i \mathbb{C}^{n+1}$.
- Exponents are $\{1, 2, \dots, n\}$.

- Group theory

- $G_{\text{sc}} = \text{SL}_{n+1}(\mathbb{C})$.
- $G_{\text{ad}} = \text{PSL}_{n+1}(\mathbb{C})$.

B Type B_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{so}(2n+1)$, preserving

$$B = \begin{bmatrix} & & 1_n \\ & 1_n & \\ & & \\ & & & 1 \end{bmatrix}.$$

- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n, 0) : a_1, \dots, a_n\}$.
- $\mathfrak{h}^*$ is spanned by the projections $e_i : \text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n, 0) \mapsto a_i$.

- Root system

root	length	root space	coroot
$e_i - e_j$	long	$E_{ij} - E_{i+n, j+n}$	$H_i - H_j$
$e_i + e_j$	long	$E_{i, j+n} - E_{j, i+n}$	$H_i + H_j$
$-e_i - e_j$	long	$E_{i+n, j} - E_{j+n, i}$	$-H_i - H_j$
e_i	short	$E_{i, 2n+1} - E_{2n+1, i+n}$	$2H_i$
$-e_i$	short	$E_{n+i, 2n+1} - E_{2n+1, i}$	$-2H_i$

Here, $H_i = E_{ii} - E_{i+n, i+n}$.

- Lattices

- Weight lattice $P = \{(\lambda_1, \dots, \lambda_n) \in (\frac{1}{2}\mathbb{Z})^n : \lambda_i - \lambda_{i+1} \in \mathbb{Z} \text{ for all } i\}$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n \geq 0\}$.
- Root lattice $Q = \mathbb{Z}^n$.
- $P/Q \cong \mathbb{Z}/2\mathbb{Z}$ by $(\lambda_1, \dots, \lambda_n) \mapsto \lambda_1$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\} \cup \{e_i + e_j : i < j\} \cup \{e_i : i\}$.
- The simple roots are $\{e_i - e_{i+1} : i\} \cup \{e_n\}$. Set $\alpha_i := e_i - e_{i+1}$ and $\alpha_n := e_n$.
- Dynkin diagram:

$$\alpha_1 \text{ --- } \alpha_2 \text{ --- } \dots \text{ --- } \alpha_{n-1} \implies \alpha_n$$

- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones for $i < n$ and $\omega_n := \frac{1}{2}(1, \dots, 1)$.
- Maximal root $\theta = e_1 + e_2$.

- Combinatorics

- Weyl group is $W = S_n \ltimes \{\pm 1\}^n$ acting by permutations of coordinates and sign changes.
- The simple reflection s_i is $(i, i+1)$ for $i < n$, and s_n is $(+1, \dots, +1, -1)$.
- $w_0 = (-1, \dots, -1)$.

- Representation theory

- $\rho = (n-1/2, n-3/2, \dots, 3/2, 1/2)$ and $\rho^\vee = (n, n-1, \dots, 2, 1)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_1\}$.
- Fundamental representations are $L_{\omega_i} = \wedge^i \mathbb{C}^{2n+1}$ for $i < n$ and the standard Clifford module for $i = n$.
- Exponents are $\{1, 3, \dots, 2n-3, 2n-1\}$.

- Group theory

- $G_{\text{sc}} = \text{Spin}_{2n+1}(\mathbb{C})$.
- $G_{\text{ad}} = \text{SO}_{2n+1}(\mathbb{C})$.

C Type C_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{sp}(2n)$, preserving

$$B = \begin{bmatrix} & -1_n \\ 1_n & \end{bmatrix}.$$

- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) : a_1, \dots, a_n\}$.
- $\mathfrak{h}^*$ is spanned by the projections $e_i : \text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) \mapsto a_i$.

- Root system

root	length	root space	coroot
$e_i - e_j$	short	$E_{ij} - E_{i+n, j+n}$	$H_i - H_j$
$e_i + e_j$	short	$E_{i, j+n} - E_{j, i+n}$	$H_i + H_j$
$-e_i - e_j$	short	$E_{i+n, j} - E_{j+n, i}$	$-H_i - H_j$
$2e_i$	long	$E_{i, i+n}$	H_i
$-2e_i$	long	$E_{i+n, i}$	$-H_i$

Here, $H_i = E_{ii} - E_{i+n, i+n}$.

- Lattices

- Weight lattice $P = \mathbb{Z}^n$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n \geq 0\}$.
- Root lattice $Q = \{(\lambda_1, \dots, \lambda_n) \in \mathbb{Z}^n : \lambda_1 + \dots + \lambda_n \equiv 0 \pmod{2}\}$.
- $P/Q \cong \mathbb{Z}/2\mathbb{Z}$ by $(\lambda_1, \dots, \lambda_n) \mapsto \sum_i \lambda_i$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\} \cup \{e_i + e_j : i < j\} \cup \{2e_i : i\}$.
- The simple roots are $\{e_i - e_{i+1} : i\} \cup \{2e_n\}$. Set $\alpha_i := e_i - e_{i+1}$ and $\alpha_n := 2e_n$.
- Dynkin diagram:

$$\alpha_1 \text{ --- } \alpha_2 \text{ --- } \dots \text{ --- } \alpha_{n-1} \text{ } \Longleftarrow \alpha_n$$

- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones.
- Maximal root $\theta = 2e_1$.

- Combinatorics

- Weyl group is $W = S_n \ltimes \{\pm 1\}^n$ acting by permutations of coordinates and sign changes.
- The simple reflection s_i is $(i, i+1)$ for $i < n$, and s_n is $(+1, \dots, +1, -1)$.
- $w_0 = (-1, \dots, -1)$.

- Representation theory

- $\rho = (n, n-1, \dots, 2, 1)$ and $\rho^\vee = (n-1/2, n-3/2, \dots, 3/2, 1/2)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_1\}$.
- Fundamental representations are $L_{\omega_i} = \ker(\iota_B : \wedge^i \mathbb{C}^{2n} \rightarrow \wedge^{i-2} \mathbb{C}^{2n})$, where ι_B is contraction:

$$\iota_B(v_1 \wedge \dots \wedge v_i) := \sum_{r < s} (-1)^{r+s+1} B(v_r \wedge v_s)(v_1 \wedge \dots \wedge \widehat{v_r} \wedge \dots \wedge \widehat{v_s} \wedge \dots \wedge v_i).$$

- Exponents are $\{1, 3, \dots, 2n-3, 2n-1\}$.

- Group theory

- $G_{\text{sc}} = \text{Sp}_{2n}(\mathbb{C})$.
- $G_{\text{ad}} = \text{Sp}_{2n}(\mathbb{C})/\{\pm 1\}$.

D Type D_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{so}(2n)$, preserving

$$B = \begin{bmatrix} & 1_n \\ 1_n & \end{bmatrix}.$$

- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) : a_1, \dots, a_n\}$.
- $\mathfrak{h}^*$ is spanned by the projections $e_i : \text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) \mapsto a_i$.

- Root system

root	root space	coroot
$e_i - e_j$	$E_{ij} - E_{j+n, i+n}$	$H_i - H_j$
$e_i + e_j$	$E_{i, j+n} - E_{j, i+n}$	$H_i + H_j$
$-e_i - e_j$	$E_{i+n, j} - E_{j+n, i}$	$-H_i - H_j$

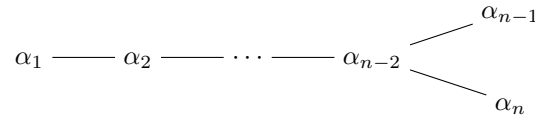
Here, $H_i = E_{ii} - E_{i+n, i+n}$.

- Lattices

- Weight lattice $P = \{(\lambda_1, \dots, \lambda_n) \in (\frac{1}{2}\mathbb{Z})^n : \lambda_i - \lambda_{i+1} \in \mathbb{Z} \text{ for all } i\}$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n, \lambda_{n-1} + \lambda_n \geq 0\}$.
- Root lattice $P = \{(\lambda_1, \dots, \lambda_n) \in \mathbb{Z}^n : \lambda_1 + \dots + \lambda_n \equiv 0 \pmod{2}\}$.
- $P/Q \cong (\mathbb{Z}/2\mathbb{Z})^2$ if n even and $P/Q \cong \mathbb{Z}/4\mathbb{Z}$ if n odd. It is generated by $(1, 0, \dots, 0)$ and $\frac{1}{2}(1, \dots, 1)$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\} \cup \{e_i + e_j : i < j\}$.
- The simple roots are $\{e_i - e_{i+1} : i\} \cup \{e_{n-1} + e_n\}$. Set $\alpha_i := e_i - e_{i+1}$ and $\alpha_n := e_{n-1} + e_n$.
- Dynkin diagram:



- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones for $i < n - 1$ and $\omega_{n-1} := \frac{1}{2}(1, \dots, 1, -1)$ and $\omega_n := \frac{1}{2}(1, \dots, 1)$.
- Maximal root $\theta = e_1 + e_2$.

- Combinatorics

- Weyl group is $W = S_n \ltimes \{\pm 1\}_+^n$ acting by permutations of coordinates and even sign changes.
- The simple reflection s_i is $(i, i + 1)$ for $i < n$, and s_n is $(+1, \dots, +1, -1, -1)$.
- If n even, $w_0 = (-1, \dots, -1)$ acts by (-1) on P/Q . If n odd, $w_0 = (-1, \dots, -1, +1)$ does not.

- Representation theory

- $\rho = (n - 1, n - 2, \dots, 1, 0)$ and $\rho^\vee = (n - 1, n - 2, \dots, 1, 0)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_1, \omega_{n-1}, \omega_n\}$.
- Fundamental representations are $L_{\omega_i} = \wedge^i \mathbb{C}^{2n+1}$ for $i < n$ and the degree pieces of the standard Clifford module.
- Exponents are $\{1, 3, \dots, 2n - 3, 2n - 1\} \sqcup \{n\}$.

- Group theory

- $G_{\text{sc}} = \text{Spin}_{2n}(\mathbb{C})$.
- $G_{\text{ad}} = \text{SO}_{2n}(\mathbb{C})/\{\pm 1\}$.